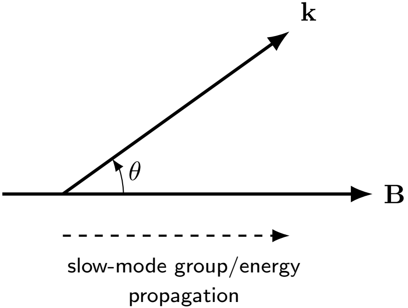

Linear magnetosonic speeds

$$v_{f,s}^2 = \frac{1}{2}\left[v_A^2 + c_s^2 \pm \sqrt{(v_A^2 + c_s^2)^2 - 4v_A^2 c_s^2 \cos^2\theta}\right]$$

Low-β limit

$v_{\mathrm{slow}} \simeq c_s|\cos\theta|$, so $v_{\mathrm{slow}} \to 0$ as $\theta \to 90^\circ$.

A simple linear slow wave therefore cannot carry a global front strictly across a uniform field.

This restriction applies to the **linear slow branch**. An oblique nonlinear slow shock is different mathematically, but it remains constrained by magnetic geometry and dissipation.